

Optimal Sampling Strategies for the Measurement of Spin–Spin Relaxation Times

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It is shown how Cramér–Rao theory may be used to determine the optimal sampling pattern for measuring the NMR spin–spin relaxation time, T_2 . The results may also be applied to the measurement of any other exponential decay, including some pulse sequences for measuring the spin–lattice relaxation time, T_1 . The optimal sampling pattern involves placing 22% of the sample points at zero time and the remaining 78% at $1.28 T_2$, or, more practically and almost as accurately, one point at zero and four at $1.30 T_2$. These sampling patterns are very different from those commonly used. The Cramér–Rao results are compared with experimental measurements and computer simulations. Some limitations of the method are described, and its extension to the simultaneous measurement of a range of T_2 values is addressed. © 1996

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INTRODUCTION

Many NMR experiments involve the measurement of relaxation times, as a valuable source of information on molecular motion, including internal motions in biological macromolecules such as proteins (1–3). Relaxation times are determined by sampling a relaxing magnetization after various delay times, and for the pulse sequences commonly used to study proteins, both spin–lattice and spin–spin relaxation result in an exponential decay of signal intensity toward zero. While much effort has been directed to improving these experiments, surprisingly little study has been made of the optimal choice of sample times [the related problem of choosing sample times in an inversion-recovery experiment has been studied in detail, as described in (4) and references therein]. Typical sampling patterns used hitherto involve 5–10 sample points, spaced approximately equally in time, together with a few points at very long times and very short times. These patterns appear to be determined largely by tradition and intuition, rather than from any more solid basis.

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Here we use the theory of Cramér–Rao lower bounds (5–8) to determine the optimum choice of sampling times for measuring T_2 in a spin-echo-type experiment. The results are not limited to T_2 measurements, but may also be applied to any experiment involving exponentially decaying signal intensity, including some sequences used to measure T_1 in proteins. Unlike previous approaches, our optimal patterns involve sampling at a small number of carefully chosen times.

Cramér–Rao lower bounds (also known as “minimum-variance bounds”) provide a fundamental limit to the accuracy with which a parameter can be determined from any given experimental data: for an unbiased data-processing method, the error in a parameter (as measured by its standard deviation) must be greater than or equal to the lower bound. In many cases, model-fitting methods result in parameters with standard deviations *equal* to the lower bounds, and so are theoretically optimal data-processing methods. This means that the optimal sampling strategy for the determination of a given parameter can be found simply by minimizing the Cramér–Rao lower bound for that parameter.

CRAMÉR–RAO LOWER BOUNDS FOR EXPONENTIAL TIME CONSTANTS

The theory (5–8) and application (9–11) of Cramér–Rao lower bounds have been well explained elsewhere, and here we only summarize their application to the measurement of exponential time constants, such as T_2 . We start with an appropriate model function

$$\hat{y}_\tau = \alpha e^{-\tau/T}, \quad [1]$$

where α is the intrinsic signal intensity, T the exponential time constant, and τ the sample time. For any parameter, θ_k , the Cramér–Rao lower bound for its standard deviation, $s(\theta_k)$, may be calculated from the Fisher information matrix, **F**, by

$$s(\theta_k) = \sqrt{(\mathbf{F}^{-1})_{kk}}. \quad [2]$$

If the experimental noise is Gaussian, with standard deviation σ , the elements of $\mathbf{F}$ are given by

$$F_{jk} = \frac{1}{\sigma^2} \sum_{\tau} \operatorname{Re} \left(\frac{\partial \hat{y}_{\tau}^*}{\partial \theta_j} \frac{\partial \hat{y}_{\tau}}{\partial \theta_k} \right). \quad [3]$$

Choosing $\theta_1 = \alpha$ and $\theta_2 = T$ gives

$$\mathbf{F} = \frac{1}{\sigma^2} \begin{pmatrix} \sum_{\tau} e^{-2\tau/T} & \sum_{\tau} \frac{\alpha\tau}{T^2} e^{-2\tau/T} \\ \sum_{\tau} \frac{\alpha\tau}{T^2} e^{-2\tau/T} & \sum_{\tau} \frac{\alpha^2\tau^2}{T^4} e^{-2\tau/T} \end{pmatrix}, \quad [4]$$

and so $s(\alpha)$ and $s(T)$ can be readily calculated.

The simplest case is when the exponential decay is sampled at two times, τ_0 and τ_1 , for which

$$s(T) = \frac{\sigma T}{\alpha} g(\tau'_0, \tau'_1) \quad [5]$$

with

$$g(\tau'_0, \tau'_1) = \sqrt{\frac{e^{2\tau'_0} + e^{2\tau'_1}}{(\tau'_1 - \tau'_0)^2}}, \quad [6]$$

where $\tau'_0 = \tau_0/T$ and similarly for τ'_1 . Several conclusions can be drawn from the form of Eq. [5]. First, the error in determining T increases with the noise level and decreases with the amplitude of the data. In other words, the error in T is inversely proportional to the signal-to-noise ratio. However, as discussed below, model fitting will break down and Cramér–Rao theory will not be applicable if the signal-to-noise ratio is too low. Second, the error in determining T is proportional to T . This is not very interesting and simply reflects the fact that the fractional error with which T may be determined is independent of T . Third, except for the dependencies discussed above, the accuracy with which T may be determined depends only on the sampling times when expressed as fractions of T . Consequently, it is possible to give general guidelines about how sampling times should be chosen, independent of the actual values of T being considered.

Given the form of Eq. [5], we prefer to describe our results in terms of the “reliability” of a parameter, defined as the ratio of a parameter’s value to the error in this value:

$$R(T) = \frac{T}{s(T)} = \frac{\alpha}{\sigma g(\tau'_0, \tau'_1)}. \quad [7]$$

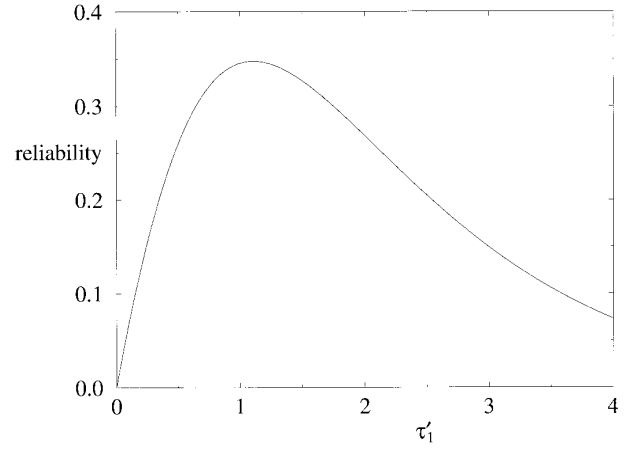


FIG. 1. Plot of the fractional reliability of T determined by sampling at two time points, $\tau_0 = 0$ and $\tau_1 = \tau'_1 T$, Eq. [10]. The maximum occurs at $\tau_1 = 1.11T$.

This reliability is simply the reciprocal of the fractional error in a parameter and is directly proportional to the signal-to-noise ratio. Furthermore, the reliability is independent of the value of T , except insofar as it affects the function g . Finally, as the signal-to-noise ratio merely scales the reliability, it is often convenient to drop this term, and use the fractional reliability

$$r(T) = \frac{R(T)}{\alpha/\sigma} \quad [8]$$

$$= \sqrt{\frac{(\tau'_1 - \tau'_0)^2}{e^{2\tau'_0} + e^{2\tau'_1}}}. \quad [9]$$

Assuming that $\tau_1 > \tau_0$, this equation may also be written

$$r(T) = e^{-\tau'_0} \frac{\tau'_1 - \tau'_0}{\sqrt{1 + e^{2(\tau'_1 - \tau'_0)}}}. \quad [10]$$

The first factor in this equation indicates that the reliability is at a maximum when the first sample point is as early as possible. This result is hardly surprising, as delaying the first point simply reduces the available signal. The second term is more interesting and shows that the reliability will initially rise as $\tau'_1 - \tau'_0$ is increased, but will eventually fall. When $\tau_0 = 0$, the maximum occurs at $\tau_1 = 1.11 T$, as shown in Fig. 1, with a maximum fractional reliability of 0.347.

This result can be easily rationalized. When $\tau_1 - \tau_0$ is small compared with T , little signal decay has occurred between the two sample points, and so the data depend only weakly on the value of T ; when $\tau_1 - \tau_0$ is large compared with T , the signal has undergone substantial decay, but this results in a low signal intensity at the second sample point. Clearly, there exists an optimum time at which the signal

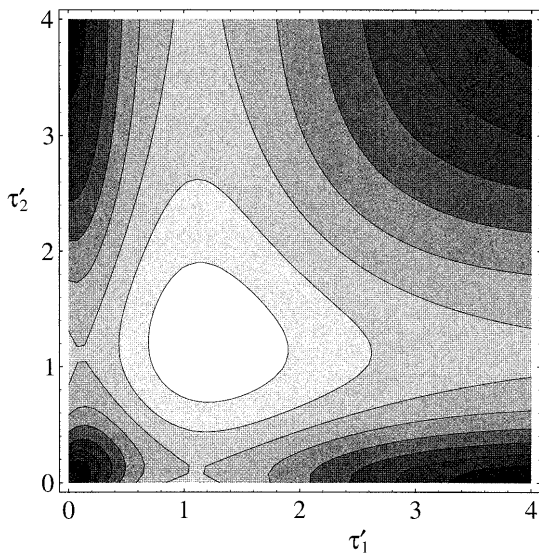


FIG. 2. Plot of the fractional reliability of T determined by sampling at three time points, 0, τ_1 , and τ_2 . The maximum occurs at $\tau_1 = \tau_2 = 1.19T$.

has relaxed sufficiently that the decay can be readily seen, but for which the signal intensity remains reasonably high.

The fractional reliability of T can be calculated in a similar way for any number of sample points. It is always advantageous to place the first point as early as possible, and in general we will assume that $\tau_0 = 0$. In cases where the first point cannot be placed at zero time, due to experimental limitations, all other sampling times should be measured from this first point. The result for three sample points at times 0, τ_1 , and τ_2 is

$$r(T) = \sqrt{\frac{(\tau'_2)^2 e^{2\tau'_1} + (\tau'_1)^2 e^{2\tau'_2} + (\tau'_1 - \tau'_2)^2}{e^{2\tau'_1} + e^{2\tau'_2} + e^{2(\tau'_1 + \tau'_2)}}}, \quad [11]$$

where $\tau'_i = \tau_i/T$ as before. This function is shown as a contour plot in Fig. 2. The global maximum occurs at $\tau_1 = \tau_2 = 1.19T$. The optimum sampling pattern involves making two measurements at the same sample time, and this is a general feature of sampling patterns for measuring exponential decay times. Of course, it is not actually necessary to record and process two *separate* samples at the second time point: the same effect can be achieved by simply making a single measurement with twice as many scans. The optimal sampling time is slightly later for three-point sampling than with two points, but this can easily be explained: making two measurements at a time point effectively increases the signal intensity at that point, allowing the measurements to be made further into the decay.

The maximum fractional reliability achievable with three points is 0.470, substantially larger than that for two-point sampling. However, this ignores the fact that using three

sample points requires more measurements, and hence a longer time. This extra time could instead be used for additional signal averaging, thus improving the signal-to-noise ratio of the data points in a two-sample-point pattern. A more useful measure is the fractional reliability divided by the square root of the number of sample points acquired, which corrects for this effect. This normalized reliability is 0.246 for two-point sampling, and 0.272 for three-point sampling, showing that the latter *does* give a genuine improvement of about 10%.

The equations for larger numbers of sampling points are complicated and difficult to depict. They can, however, be explored by evaluating the reliability at a large number of different sets of sample points, evenly spaced on a grid, and the results are easily summarized. For any number of sample points, it is *always* best to place some samples at $\tau = 0$ and the remainder at some optimal time, τ_{opt} . The results for some specific cases are given in Table 1. When the number of sample points, N_{total} , is very large, the optimal pattern is to place 22% of the points at $\tau = 0$, and the remaining 78% at $\tau = 1.28T$. Note that this optimal sampling pattern gives the same total signal intensity at the two sample times ($0.22 = 0.78e^{-1.28}$); this point is discussed more fully in the Appendix of Ref. (11). When N_{total} is small, the optimal pattern approximates this ideal pattern and is slightly less efficient, where the efficiency is defined as the normalized reliability, $r(T)/\sqrt{N_{\text{total}}}$, measured relative to the optimal experiment with an infinite number of samples. The sampling pattern for $N_{\text{total}} = 5$ (one point at $\tau = 0$ and four points at $\tau = 1.30T$) has almost identical efficiency to the ideal pattern, and for all practical purposes can be treated as ideal.

Our optimal sampling strategies are somewhat unconventional, and it is instructive to consider whether more traditional sampling patterns might have any advantages. The optimal strategies described above do suffer from two potentially serious weaknesses. First, they *assume* that the observed decay is indeed well described by a single exponen-

TABLE 1
Variation of Optimal Sampling Pattern for Measuring T with the Number of Sample Points (N_{total})

N_{total}	Sampling pattern ^a	Efficiency ^b
2	$\tau = 0$ (1), $\tau = 1.11 T$ (1)	0.882
3	$\tau = 0$ (1), $\tau = 1.19 T$ (2)	0.975
4	$\tau = 0$ (1), $\tau = 1.25 T$ (3)	0.998
5	$\tau = 0$ (1), $\tau = 1.30 T$ (4)	0.999
6	$\tau = 0$ (1), $\tau = 1.34 T$ (5)	0.993
7	$\tau = 0$ (2), $\tau = 1.22 T$ (5)	0.991
∞	$\tau = 0$ (1), $\tau = 1.28 T$ (3.59)	1.000

^a Numbers in parentheses give the number of sample points at each time.

^b Defined as the normalized reliability, $r(T)/\sqrt{N_{\text{total}}}$, measured relative to the optimal $N_{\text{total}} = \infty$ experiment.

tial, and it is impossible to judge this assumption using only two points. Second, the patterns are only optimal if the true value of T is close to the value assumed when the sampling times were derived. Both these problems are addressed in more detail under Discussion below.

EXPERIMENTAL VERIFICATION

To verify the predictions above, we undertook an extensive series of relaxation measurements. By recording the same data many times, it is possible to study the distribution of parameters obtained, and so to estimate the reliability of a parameter. As relaxation measurements are important in the study of proteins, our experiments were designed to emulate ^{15}N relaxation measurements in peptides and proteins.

^{15}N relaxation times in proteins are typically measured using a series of modified ^1H ^{15}N heteronuclear single-quantum-coherence (HSQC) experiments (1). The HSQC sequence is modified by the addition of a relaxation period before the ^{15}N evolution period, resulting in a two-dimensional spectrum whose peak intensities are weighted by the appropriate relaxation process. For the pulse sequences commonly used, both spin-lattice relaxation (T_1) and spin-spin relaxation (T_2) result in an exponential decay of signal intensity to zero, and so the above predictions are applicable.

For relaxation studies of proteins, the two-dimensional HSQC experiment is used to overcome the lack of resolution in the ^1H dimension. This greatly increases the acquisition time of the experiment, making it impractical to acquire many spectra. For small peptides, the resolution in the ^1H dimension is adequate, and so it is not necessary to acquire a full two-dimensional experiment; instead, a simple one-dimensional HSQC-filtered experiment will suffice. This allows a much larger number of experiments to be recorded in a reasonable time.

As a test molecule, we chose to use the tripeptide glutathione ($\gamma\text{-Glu-Cys-Gly}$). An approximately 1 M solution was prepared by adding 150 mg of glutathione to 0.5 ml of 95% $\text{H}_2\text{O}/5\%$ D_2O , and adjusting the pH to approximately 5. At this pH, the two peptide-bond NH protons are in slow exchange, while the terminal amino protons are in fast exchange; at higher pH, the peptide-bond NH protons also exchange rapidly, while at lower pH, the peptide has much lower solubility. At 1 M solution is approximately 3 mM in ^{15}N , comparable to the concentration used in studies of fully ^{15}N -labeled proteins. The sample was partly deoxygenated by bubbling nitrogen gas through the solution in the NMR tube; under these conditions, the sample is stable for several days. Experiments were carried out at 17°C using a home-built spectrometer, with a ^1H frequency of 600.2 MHz. ^{15}N spin-spin relaxation was studied using a modified CPMG sequence (12), incorporating 180° pulses on protons to average the relaxation rates of the two components of each NH doublet. Field-gradient coherence pathway selection was

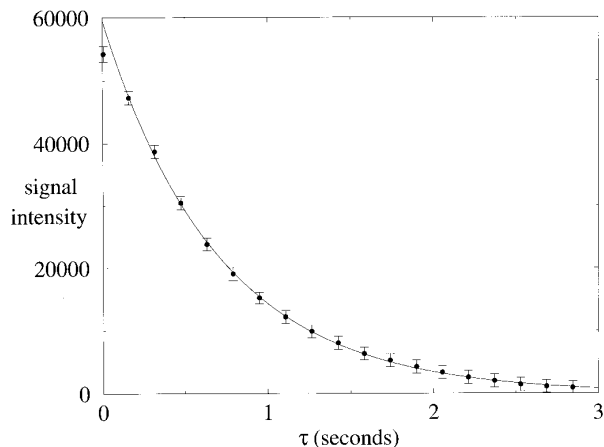


FIG. 3. Plot of the integrated intensity (arbitrary units) of the cysteine NH resonance as a function of relaxation time. The data points show mean $\pm$ standard deviation, while the smooth curve is an exponential decay fitted to all but the first point. 256 measurements were made at each time point.

used to suppress signals which did not pass through the HSQC filter (13–15). An acceptable signal-to-noise ratio was obtained after four scans.

Relaxation measurements were made using 19 different relaxation delays, at integral multiples of 158 ms (corresponding to 50 repetitions of the CPMG pulse train). Each relaxation measurement was repeated 256 times, and the results were stored separately. The total measurement time was about 20 hours. Each data set was then weighted, Fourier transformed, and phased using the program FELIX (Hare Research). The resulting spectra contained two multiplets, corresponding to the two peptide-bond NH protons. As the two signals showed very similar relaxation behavior, only the low-field multiplet (arising from the NH group of cysteine) was processed further. A plot of integrated intensity against decay time is shown in Fig. 3. It is clear that the first time point does not lie on the main exponential decay, and so this point was not used in later processing. Careful examination of the other data points showed that the observed decay was not perfectly exponential, but deviations from this ideal behavior were small.

The remaining 18 time points were analyzed by model fitting to obtain estimates of the relaxation time, T_2 , and the initial amplitude, α ; this analysis was repeated using different numbers and combinations of time points. In general, these parameters were estimated using χ^2 minimization, based on the SPIRAL routine (16), but when only two time points are used, model fitting can be achieved analytically. Initially, the data were analyzed by fitting to all 18 sample times, giving 256 values of α and T_2 . The mean and standard deviation were calculated for each parameter, giving the values shown in Table 2. The mean values for α and T_2 thus obtained were then used as best estimates for these parameters when calculating predicted reliabilities.

TABLE 2
Relaxation Parameters Obtained by Model Fitting
to All 18 Time Points

Parameter	Mean	Standard deviation
α^a	596500	12900
T_2^b	0.7009	0.0202

^a Arbitrary units.

^b Measured in seconds.

In order to test the validity of Eq. [7], the value of T_2 was estimated using the first sample time and one other. The results are plotted in Fig. 4. Reliabilities and the errors in the reliabilities were estimated using the Monte Carlo bootstrap method (see Appendix). For most choices of sample time, the experimental data points lie pleasingly close to the theoretical Cramér–Rao curve, but for values of τ_1 which are very small or very large, the experimental measurements are less reliable than predicted. This discrepancy occurs because for these time points model fitting begins to break down, and so the simple analytic theory is no longer applicable. The behavior in these regions can only be explored by computer simulations (17, 18). Essentially, this involves repeating the experiment described above, but using “data” synthesized according to the estimated values of α and T_2 and the noise distribution. As the data are synthetic, very large data sets can be produced; the dashed curve drawn in Fig. 4 corresponds to 10^6 data sets for each sample point. The experi-

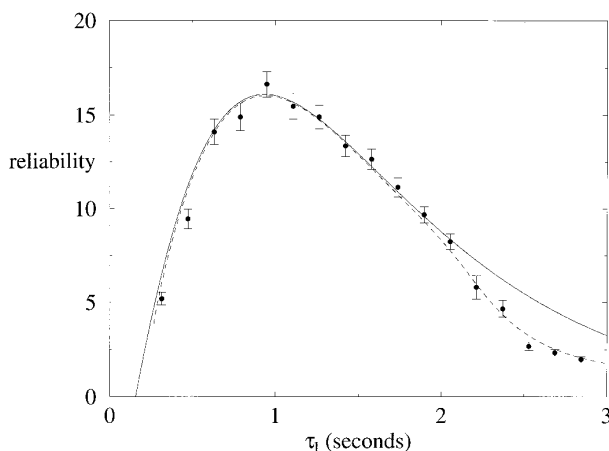


FIG. 4. Reliability of T_2 obtained by fitting to the first sample time ($\tau_0 = 158$ ms) and one other (τ_1). The data points show reliability $\pm$ uncertainty, calculated using the Monte Carlo bootstrap method (19, 20). The solid line shows the theoretical Cramér–Rao curve, calculated using Eq. [7] with values of α and T taken from Table 2. The dashed line shows the curve estimated by computer Monte Carlo simulations, using 10^6 data sets at each sampling point. The Monte Carlo curve is not drawn for very early sampling times, as the calculations were not well behaved in this region.

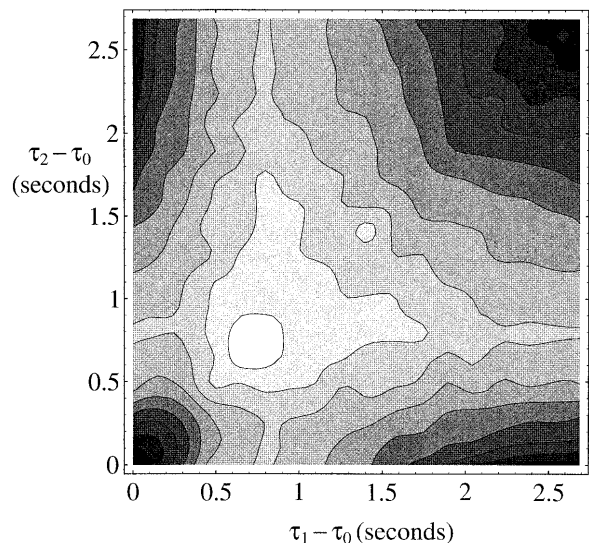


FIG. 5. Reliability of T_2 obtained by fitting to the first sample time ($\tau_0 = 158$ ms) and two others (τ_1 and τ_2). The data set was interpolated using bicubic splines before contouring. The observed maximum reliability occurs close to the predicted position ($\tau_1 - \tau_0 = \tau_2 - \tau_0 = 0.83$ s).

mental measurements lie close to this Monte Carlo curve at all sampling times. This excellent agreement confirms both the basic validity of Eq. [7], and the expected breakdown at low signal-to-noise levels.

Similarly, the result for three sampling times can be tested by estimating the value of T_2 using the first sample time and two others. Care must be taken when the two sampling times are the same: it is not correct to use the corresponding value from a data set twice, as this does not result in two independent estimates of the signal intensity. Instead, it is necessary to take data points from two different data sets. The experimental reliability is shown as a contour plot in Fig. 5, which should be compared with Fig. 2; it is clear that the two plots have similar forms. The observed maximum reliability occurs close to the predicted position of $\tau_1 - \tau_0 = \tau_2 - \tau_0 = 0.83$ s.

A similar correlation exists between the experimental reliability and predicted Cramér–Rao reliability for larger numbers of sample times. Figure 6 shows the results of estimating the reliability using the first sample time and any four other times (all chosen to be different from one another). There are 2380 possible choices of sampling times, which makes any detailed comparison difficult, but it is clear that the experimental reliability is generally quite similar to that predicted by Cramér–Rao theory. The observed scatter is partly due to the limited signal-to-noise ratio, and partly due to the breakdown of Cramér–Rao theory for some sampling patterns.

OPTIMAL LINEAR SAMPLING SCHEMES

The optimal sampling patterns described above are very different from those commonly used. A less extreme ap-

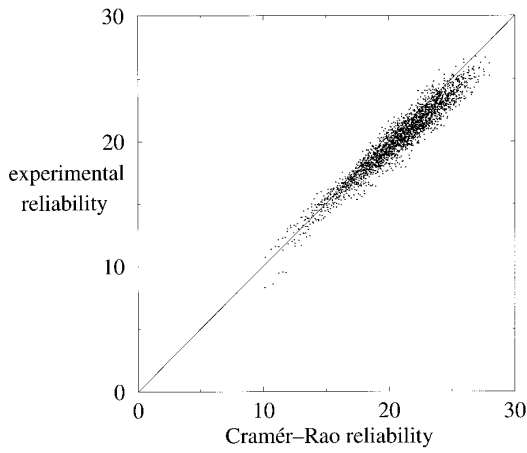


FIG. 6. Reliability of T_2 obtained by fitting to the first sample time and any four others. The experimental reliability is plotted against the theoretical Cramér–Rao reliability. The solid line has a gradient of one.

proach involves taking a family of more conventional sampling schemes and finding the optimum member. This scheme will, of course, be less efficient than the true optimal sampling pattern, but may have other advantages.

Linear sampling is a particularly simple family of sampling schemes. In this case, sample times are placed evenly between zero and some maximum time, $\tau_{\max}$. The fractional reliability achievable using such a scheme will depend only on the number of sample points, N_{total} , and $\tau_{\max}$ expressed as a multiple of T . For example, for $N_{\text{total}} = 3$ (three sample points at 0, $\tau_{\max}/2$, and $\tau_{\max}$),

$$r(T) = \frac{\tau_{\max}/T}{2e^{\tau_{\max}/2T}} \sqrt{\frac{1 + 4e^{\tau_{\max}/T} + e^{2\tau_{\max}/T}}{1 + e^{\tau_{\max}/T} + e^{2\tau_{\max}/T}}} \quad [12]$$

This equation can be maximized numerically. The optimal three-point linear sampling pattern turns out to involve sampling at 0, $0.80T$, and $1.60T$, with a normalized reliability of 0.253. The results for other values of N_{total} are shown in Table 3.

TABLE 3

Variation of Optimal Linear Sampling Pattern for Measuring T with the Number of Sample Points (N_{total})

N_{total}	$\tau_{\max}^a$	Efficiency ^b
2	1.11 T	0.882
3	1.60 T	0.909
4	1.82 T	0.892
5	1.93 T	0.872
6	1.99 T	0.855
7	2.01 T	0.841
∞	2.02 T	0.742

^a Sample times are evenly spaced between 0 and $\tau_{\max}$.

^b Defined as the normalized reliability, $r(T)/\sqrt{N_{\text{total}}}$, measured relative to the optimal $N_{\text{total}} = \infty$ experiment.

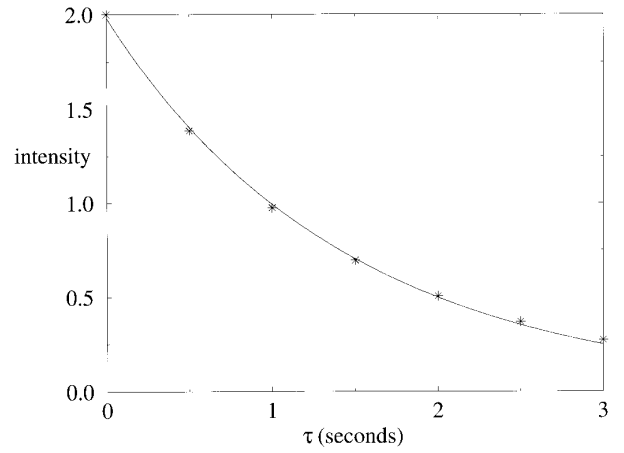


FIG. 7. Plot of a biexponential decay. The data points were calculated as the sum of two exponentials with the same amplitude, and time constants of 1 and 2 s, respectively. The smooth curve was obtained by fitting a single-exponential decay to the seven data points, and lies very close to them. The systematic deviations arising from the assumption of an incorrect model function will be overwhelmed by random errors unless the signal-to-noise ratio is high (at least 100).

As expected, the linear sampling schemes are generally less efficient than the optimal sampling schemes with the same number of sampling points (the one exception being $N_{\text{total}} = 2$, for which the two schemes are identical). However, as long as N_{total} is reasonably small (less than 7), the loss in efficiency is no more than 15%, which may be acceptable in practice. Linear sampling schemes with very large numbers of points have a substantially lower efficiency and should be avoided.

DISCUSSION

Cramér–Rao theory provides a convenient framework for discussing the optimum selection of sampling times, and as the results above demonstrate, its predictions are confirmed by experiment as long as the signal-to-noise ratio is reasonably high. When this is not the case, the Cramér–Rao predictions are optimistic; that is, experiments will generally do worse than expected. Nevertheless, the general form of the reliability and the optimal choice of sampling times are still predicted fairly well.

The optimal sampling patterns derived from Cramér–Rao theory involve sampling at only two times, and this exposes them to a major problem: it is impossible to test the underlying assumption that the signal decay is exponential. With sample patterns containing three or more sample points, it is *theoretically* possible to detect any deviations from an exponential decay. Such deviations can have many causes, including imperfections in the CPMG sequence, and overlap of peaks with different relaxation times. In practice, however, it is very difficult to detect small deviations from exponential behavior, as shown in Fig. 7. Unless the components

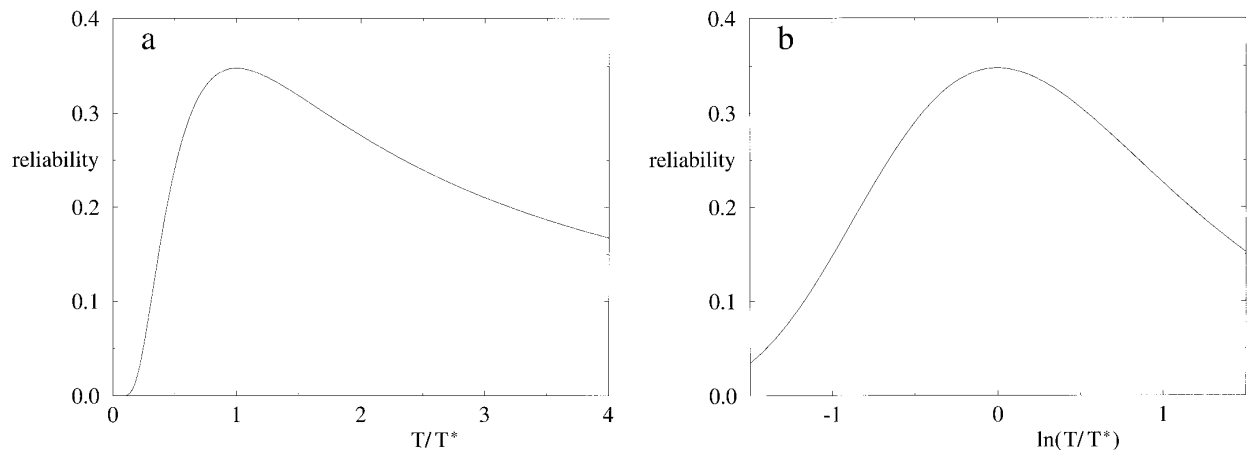


FIG. 8. Fractional reliability of T obtained by fitting to two sample times. Dependence of this reliability on the ratio between T and the assumed relaxation time T^* for which the sampling pattern has been optimized: (a) plotted as a function of T/T^* , and (b) plotted as a function of $\ln(T/T^*)$.

of the decay have time constants differing by more than a factor of two, it is effectively impossible convincingly to detect a biexponential decay at realistic signal-to-noise levels. Major deviations from a simple exponential decay can cause serious problems, but are unlikely to arise, with one exception: signals that consist of the sum of an exponential decay and a constant offset. This situation can arise when the signal intensity contains a component created by pulses after the decay period, as this component will be insensitive to the choice of decay times. This potential problem is, however, largely solved by the widespread use of field gradients to select coherence-transfer pathways (13–15).

A second major problem with *any* optimized sampling scheme is that the value of T must be known before the best sampling pattern for estimating T can be determined. While it is normal to have some idea of the value of T before starting the experiment, it is important to investigate how well the method copes with errors in this initial estimate. In general, the true value of T will lie in some range around the estimated value, T^* , and a sampling pattern is of little use if it is not fairly efficient throughout this range. A related, and ultimately equivalent, problem concerns the optimal sampling pattern when measuring several resonances with different T values: it is important to choose a sampling pattern which is reasonable for all values of T in the range, and not just some of them.

This question is addressed in Fig. 8, which shows the reliability of a T measurement using two sample times as a function of the ratio of the actual T value to the assumed value, T^* . It is clear that the reliability remains fairly high as long as T does not greatly differ from T^* , but for more substantial deviations the reliability can be drastically reduced. Furthermore, it is clear from Fig. 8a that the sampling pattern performs much better for $T > T^*$ than for $T < T^*$. This suggests that it may be best to optimize the sampling pattern for a value T^* slightly smaller than the best estimate

of T . Similarly, if it is necessary to measure a range of T values, it might seem sensible to optimize the sampling pattern for a value of T^* a little smaller than the mean of the range. Choosing the mean itself will give reliable results for T values close to or larger than this mean value, but will perform poorly for T values much smaller than the mean value. A better approach is suggested by Fig. 8b, which shows the same plot with a logarithmic T axis. This graph is much more symmetric than Fig. 8a.

An exactly symmetric graph would imply that a sampling pattern optimized for $T = T^*$ would perform equally well for $T = T^*f$ and $T = T^*/f$. If this were true, it would be simple to determine the optimal two-point sampling pattern for a range of T values: first, calculate the *geometric* mean of the extreme values of the range

$$T^{\text{GM}} = \sqrt{T^{\text{min}}T^{\text{max}}} \quad [13]$$

and then optimize the sampling pattern for T^{GM} . In fact, the graph is not perfectly symmetric, but falls more rapidly for $T < T^*$. This means that the simple prescription above is not quite correct, and $R(T^{\text{min}})$ will be slightly lower than $R(T^{\text{max}})$. A better choice would be to use a value of T^* slightly smaller than T^{GM} , thus increasing $R(T^{\text{min}})$ and decreasing $R(T^{\text{max}})$. For the ideal choice (T^{opt}), $R(T^{\text{min}}) = R(T^{\text{max}})$: this ideal value can be found numerically.

The results of using this approach are plotted in Fig. 9. The optimal two-point sampling pattern is to place one point at 0 and the other at $1.11T^{\text{opt}}$, where T^{opt} is determined as described above. The solid line shows the ratio $T^{\text{opt}}/T^{\text{GM}}$ as a function of the width of the range of T values, $T^{\text{max}}/T^{\text{min}}$. The dashed line shows the reliability which can be achieved at the extremes of the range (T^{max} and T^{min}), measured as a fraction of the reliability at T^{opt} . For values of T within the chosen range, the reliability will always be higher than

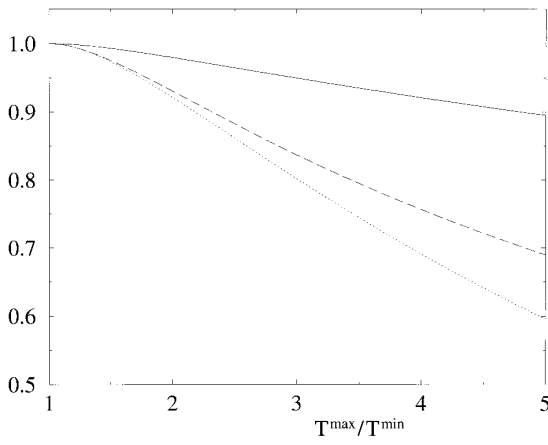


FIG. 9. Optimum two-point sampling pattern for a range of T values plotted as a function of $T^{\max}/T^{\min}$. The optimum pattern comprises one point at 0 and one at $1.11T^{\text{opt}}$. The solid line shows $T^{\text{opt}}/T^{\text{GM}}$, where T^{GM} is the geometric mean of $T^{\min}$ and $T^{\max}$. The dashed line shows the reliability at the extremes of the range, measured as a fraction of the reliability at T^{opt} . The dotted line shows the reliability at $T^{\min}$ if it is assumed (incorrectly) that $T^{\text{opt}} = T^{\text{GM}}$.

for these worst-case values. The dotted line shows the effect of incorrectly assuming that $T^{\text{opt}} = T^{\text{GM}}$; in this case, the reliability at $T^{\min}$ will be reduced to the value shown.

The significance of this figure can be clarified by a few examples. First, consider a fairly narrow range of T values, such as $T^{\max} = 2T^{\min}$. In this case, T^{GM} equals $1.41T^{\min}$, while T^{opt} is very slightly smaller ($T^{\text{opt}} = 0.98T^{\text{GM}} = 1.38T^{\min}$). Measurements of T will be most reliable when $T = T^{\text{opt}}$, and least reliable at $T^{\max}$ and $T^{\min}$, where the reliability will be only 93% of that achievable at T^{opt} . If it is assumed (incorrectly) that $T^{\text{opt}} = T^{\text{GM}}$, the maximum reliability will now be achieved at $T = T^{\text{GM}}$, and the minimum reliability, which occurs at $T = T^{\min}$, is reduced to 92%. For narrow ranges, such as this, the sampling pattern is satisfactory: the simple assumption that $T^{\text{opt}} = T^{\text{GM}}$ is adequate, and T can be measured fairly well over the whole range.

For much broader ranges of T values, the behavior is different. The reliability at the edges of the range is substantially smaller than that at the optimal value, and T^{opt} is significantly smaller than T^{GM} . For example, when $T^{\max} = 5T^{\min}$, the reliability achievable at the extreme values is only 69% of that at the optimal value. To achieve even this reliability, it is important to use $T^{\text{opt}} = 0.90T^{\text{GM}}$; using T^{GM} gives a worst case reliability of less than 60%. For such wide ranges, it is useful to consider other sampling patterns, such as the linear pattern discussed below.

The effect of a range of T values can be calculated for any sampling pattern and the results displayed much as above. In general, however, the results can become extremely complicated, and so only two more patterns will be discussed in detail: the optimal five-point sampling pattern, which is close

to the ideal sampling pattern for a single T value, and the optimal five-point linear-sampling pattern. These are good general purpose sampling patterns for narrow and wide distributions of T values respectively.

The optimal five-point sampling pattern comprises one point at 0 and four points at $1.30T$, and so may be described as a “sparse”-sampling pattern, while the optimal five-point linear-sampling pattern has five points equally spaced between 0 and $1.93T$. The reliabilities of these patterns are plotted in Fig. 10 as a function of the ratio of the actual value of T to the assumed value T^* (as before, a logarithmic scale has been used on the T axis). When T lies close to T^* , the sparse-sampling pattern gives more reliable results than the linear-sampling pattern, but the linear pattern remains fairly efficient over a much wider range, and so gives better results when T is very different from T^* . As before, the curves are not perfectly symmetric, and so the minimum reliability can be increased by using some optimized value, T^{opt} , instead of T^{GM} .

This result is shown in a different way in Fig. 11, which depicts the worst-case fractional reliability which can be achieved with five-point sampling as a function of the width of the range of T , defined as $T^{\max}/T^{\min}$. For narrow ranges of T values ($T^{\max} \leq 2T^{\min}$), the sparse-sampling pattern (solid line) is better than the linear-sampling pattern (dotted line), and $T^{\text{opt}} \approx T^{\text{GM}}$. For slightly wider ranges ($2T^{\min} \leq T^{\max} \leq 3T^{\min}$), the sparse-sampling pattern remains better, but it becomes important to use T^{opt} instead of T^{GM} (dashed line). For even wider ranges ($T^{\max} \geq 3T^{\min}$), the linear-sampling pattern optimized for T^{GM} is superior (for the linear-sampling scheme, little improvement is obtained by using T^{opt} instead of T^{GM}).

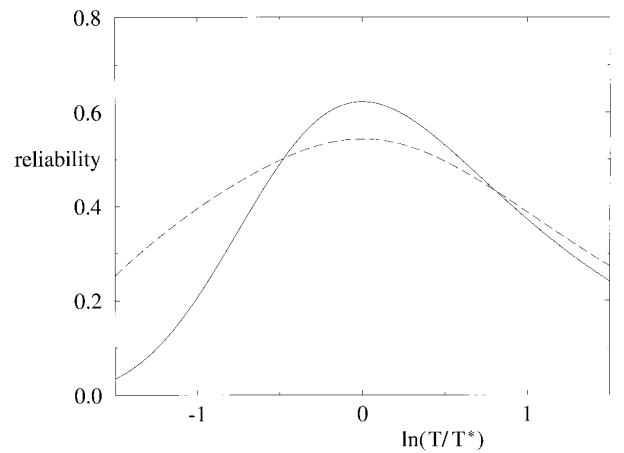


FIG. 10. Fractional reliability of T obtained by fitting to five sample times. Dependence of this reliability on $\ln(T/T^*)$ where T^* is the relaxation time for which the sampling pattern has been optimized. The solid line shows the reliability for the sparse five-point sampling pattern, while the dashed line shows the reliability for the optimal five-point linear-sampling pattern.

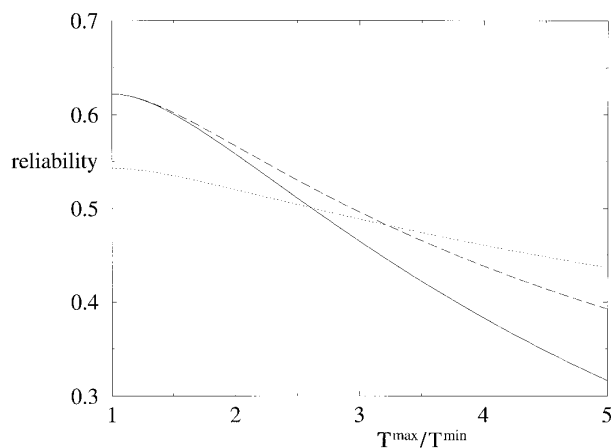


FIG. 11. The worst-case fractional reliability of five-point sampling patterns for a range of T values, plotted as a function of $T^{\max}/T^{\min}$. The solid line shows the minimum reliability achieved using one point at 0 and four points at $1.30 T^{\text{GM}}$. This result can be improved, as shown by the dashed line, by placing the four sample points at a slightly earlier time, $1.30 T^{\text{opt}}$. The dotted line shows the effect of sampling using five points linearly spaced between 0 and $1.93 T^{\text{GM}}$.

It is, of course, possible to consider additional sampling patterns, and in general, such patterns will perform better than the simple patterns described above when the range of T values is large. Unfortunately, the number of possible sampling schemes is so large that it is difficult to give any general guidance as to which is best. For a particular range of T values, it may be possible to find good sampling schemes by a brute-force search: for example, when $T^{\max} = 5T^{\min}$, a five-point sampling scheme involving one point at 0, two points at $0.6T^{\text{GM}}$, and two points at $2.3T^{\text{GM}}$ is particularly effective, with a minimum fractional reliability of 0.466. This scheme, however, is less efficient than the five-point sparse-sampling scheme unless $T^{\max} \geq 3T^{\min}$, and is less efficient than the linear scheme unless $T^{\max} \geq 4T^{\min}$. Even when $T^{\max} = 5T^{\min}$, this pattern only gives an improvement of 6% over the optimal linear-sampling scheme, and the computational complexity involved may be hard to justify.

It is useful to consider the range of time constants which may be reasonably expected in practice. Of course, this depends very strongly on the parameter being measured and the system being studied. Nevertheless, it is possible to make some general comments on the variability of ^{15}N relaxation times in globular proteins. As a specific example, we consider the ^{15}N relaxation of native hen egg-white lysozyme (12). The reported values of T_1 for this system, at 35°C and a ^1H frequency of 500 MHz, vary between 0.388 and 0.617 s, a range of less than 1.6, while the reported values of T_2 vary between 0.093 and 0.205 s, a range of about 2.2. If the smallest and largest values of T_2 are neglected, the values only extend from 0.116 to 0.189 s, a range of just over 1.6. Hence for this system, it would be sensible to use sparse

five-point sampling patterns optimized for T values of 0.489 s for T_1 and 0.148 s for T_2 (neglecting the two extreme values).

Such small ranges of T_1 and T_2 values are frequently seen in compact well-ordered proteins, suggesting that the five-point sparse-sampling pattern may be useful in many practical cases. Much larger variability is, however, seen in ^{15}N relaxation times of less-ordered molecules, such as denatured proteins, and in such cases, the linear-sampling pattern might be more appropriate.

SUMMARY

Cramér–Rao lower bounds provide a convenient method for quantifying the random errors in parameters determined from experimental data. Attempting to minimize these errors is a simple and effective way of finding sampling patterns which allow the parameters to be determined accurately.

When applied to the measurement of exponential time constants, Cramér–Rao theory predicts that the optimal sampling pattern is extremely sparse, involving sampling at only two different sample times. Twenty-two percent of the measurements should be made at 0, and the remaining 78% at $1.28T$, where T is the best prior estimate of the time constant. More practically, very similar accuracy can be obtained by placing one sample point at 0 and four at $1.30T$.

Since the optimal sampling patterns require a prior estimate of T , their performance will be degraded if there is substantial uncertainty in the value of T , or (equivalently) if the method is used simultaneously to estimate several different values of T within a range. As long as the range is narrow, the optimal sampling patterns are fairly efficient. For maximum reliability over the whole range, the sampling pattern should be optimized for a time constant slightly smaller than the geometric mean of the two extreme values (Fig. 9). If the range of T values is substantial (say, more than a factor of 3), the performance of the optimal sparse-sampling pattern is significantly degraded, and more-complex sampling patterns may be useful. While the truly optimal sampling scheme for a wide range of values of T is in general difficult to determine, a linear-sampling scheme involving five points evenly spaced between 0 and $1.93T^{\text{GM}}$ (where T^{GM} is the geometric mean of the extremes of the range as before) is usually quite effective.

Relaxation studies of biological macromolecules generally produce spectra with a low signal-to-noise ratio. Furthermore, the relaxation times frequently show little variation, and so accurate measurements are necessary if patterns in relaxation behavior are to be detected. In order to obtain the most reliable values of relaxation times, it is crucial to optimize the experiment in *every* respect. Just as significant improvements can be obtained by careful attention to sample preparation and to details of the pulse sequence, substantial

gains can also be made by a careful choice of sampling times: any other approach is a waste of precious signal.

APPENDIX

The Monte Carlo Bootstrap Method

Suppose we are attempting to estimate some parameter, θ , and have a set of N estimates of its value, θ . The reliability of θ can be estimated as follows. First calculate the sample mean, $m(\theta)$, and standard deviation, $s(\theta)$. These sample statistics can be used as reasonable estimates of the population mean, $\mu(\theta)$, and standard deviation, $\sigma(\theta)$, and so a reliability can be estimated by

$$R(\theta) = \frac{\mu(\theta)}{\sigma(\theta)} \approx \frac{m(\theta)}{s(\theta)}. \quad [\text{A.1}]$$

While this procedure provides a reasonable estimate of the reliability, it does not provide any information about how good the estimate is. Ideally, in addition to an estimate of the reliability, we would also like to have an estimate of the uncertainty in this value.

The Monte Carlo bootstrap method (19, 20) is one of several techniques, known as resampling methods (20), which attempt to address this problem. The simplest technique to describe is the "jackknife method" (20). Starting from the original data set, θ , it is possible to form N data sets, θ_i , of size $N - 1$, by deleting the i th data point. Each of these new data sets can then be used to estimate the reliability of θ , giving a set of estimates $R(\theta)$. The standard deviation of these estimates can be used as an estimate of the uncertainty in $R(\theta)$.

The Monte Carlo bootstrap method is similar, but slightly more complex. In general, it is technically superior to the jackknife method and is probably the best general purpose resampling method. It works by forming a large number, P , of data sets, θ_i , of size N , by random sampling with replacement. That is, each data point in θ_i is selected at random from θ , with no restriction on using a given data point more than once. In general, there are N^N such data sets, and the P data sets chosen form a random sample from among them. Once again each data set can be used to estimate the reliability of θ , and the standard deviation of these estimates is used as an estimate of the uncertainty in $R(\theta)$.

Unlike the jackknife method, the Monte Carlo bootstrap method has an additional adjustable parameter, the number of resampled data sets, P . In general, the results are not very sensitive to the value of P as long as it is sufficiently large. The calculations in this paper used $P = 1000$, which is large enough to give reasonable results.

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REFERENCES

1. L. E. Kay, D. A. Torchia, and A. Bax, *Biochemistry* **28**, 8972 (1989).
2. G. M. Clore, P. C. Driscoll, P. T. Wingfield, and A. M. Gronenborn, *Biochemistry* **29**, 7387 (1990).
3. J. W. Peng and G. Wagner, *Biochemistry* **31**, 8571 (1992).
4. G. H. Weiss and J. A. Ferretti, *Prog. NMR Spectrosc.* **20**, 317 (1988).
5. A. van den Bos, in "Handbook of Measurement Science" (P. H. Sydenham, Ed.), Wiley, New York, 1982.
6. J. P. Norton, "An Introduction to Identification," Academic Press, San Diego, 1986.
7. R. de Beer, D. van Ormondt, W. W. F. Pijnappel, and J. W. C. van der Ween, in "Maximum Entropy and Bayesian Methods" (J. Skilling, Ed.), Kluwer Academic, Dordrecht/Norwell, Massachusetts, 1989.
8. Y. Manassen and G. Navon, *J. Magn. Reson.* **79**, 291 (1988).
9. W. W. F. Pijnappel, Ph.D. thesis, University of Delft, 1991.
10. P. Hodgkinson, K. J. Holmes, and P. J. Hore, *J. Magn. Reson. A* **120**, 18 (1996).
11. P. Hodgkinson and P. J. Hore, *Adv. Magn. Opt. Reson.*, in press.
12. M. Buck, J. Boyd, C. Redfield, D. A. MacKenzie, D. J. Jeenes, D. B. Archer, and C. M. Dobson, *Biochemistry* **34**, 4041 (1995).
13. A. L. Davis, J. Keeler, E. D. Laue, and D. Moskau, *J. Magn. Reson.* **98**, 207 (1992).
14. J. Boyd, N. Soffe, B. John, D. Plant, and R. Hurd, *J. Magn. Reson.* **98**, 660 (1992).
15. L. E. Kay, P. Keifer, and T. Saarinen, *J. Am. Chem. Soc.* **114**, 10663 (1992).
16. A. Jones, *Computer J.* **13**, 301 (1970).
17. H. Barkhuijsen, R. de Beer, and D. van Ormondt, *J. Magn. Reson.* **67**, 371 (1986).
18. J. A. Jones and P. J. Hore, *J. Magn. Reson.* **92**, 363 (1991).
19. W. H. Press, S. A. Teukolsky, W. T. Vetterling and B. P. Flannery, "Numerical Recipes in C," 2nd ed., Cambridge Univ. Press, Cambridge, UK, 1992.
20. B. Efron, "The Jackknife, the Bootstrap and Other Resampling Plans," SIAM, Philadelphia, 1982.